

MEKNES, Morocco

WMO: 601500

Lat: 33.879N

Lon: 5.515W

Elev: 1890

StdP: 13.72

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	36.7	38.9	24.6	20.1	53.0	28.0	23.6	52.0	21.5	51.9	19.0	51.5	6.4	160	0.414

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	26.6	103.0	68.7	98.8	68.8	95.1	68.3	74.4	92.2	72.3	90.0	70.8	88.5	8.8	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.4	111.9	84.7	66.5	104.7	81.0	64.8	98.6	77.9	39.2	92.4	37.3	90.2	35.8	88.1	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.0	16.3	14.1	DB	32.1	108.5	3.1	3.0	29.9	110.6	28.1	112.4	26.3	114.1	24.1	116.3
			WB	29.6	77.3	3.4	4.6	27.2	80.6	25.2	83.3	23.3	85.8	20.9	89.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.4	51.1	52.5	56.9	60.2	66.7	73.9	78.7	79.3	73.2	67.7	58.5	53.0	(6)
	DBStd	11.45	4.53	5.02	5.54	5.57	7.15	6.75	6.34	5.92	5.42	5.98	6.18	4.49	(7)
	HDD50	102	40	30	7	0	0	0	0	0	0	0	4	21	(8)
	HDD65	1901	432	351	257	165	63	4	0	0	3	39	213	374	(9)
	CDD50	5347	73	99	222	307	516	716	888	909	697	550	258	112	(10)
	CDD65	1668	0	0	7	21	114	269	423	444	250	124	16	0	(11)
	CDH74	19259	1	8	124	342	1434	3180	5202	5342	2355	1130	138	3	(12)
	CDH80	9743	0	1	19	87	650	1632	2938	2984	1029	385	18	0	(13)
(14) Wind	WSAvg	6.4	6.4	6.7	7.1	6.9	6.5	6.4	5.6	5.8	5.9	6.0	6.5	6.6	(14)
(15) Precipitation	PrecAvg	22.4	2.9	2.8	2.7	2.3	1.5	0.5	0.1	0.1	0.7	1.9	2.8	3.0	(15)
	PrecMax	40.1	9.4	8.9	7.0	8.7	5.3	2.3	0.8	0.5	5.0	7.7	7.2	10.3	(16)
	PrecMin	13.7	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	6.5	2.4	2.0	1.9	1.9	1.3	0.6	0.2	0.1	0.9	1.7	1.9	2.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.9	73.8	82.5	87.5	98.7	104.4	107.8	107.4	100.5	91.8	82.6	73.2	(19)
		MCWB	54.3	56.9	61.1	62.5	65.0	67.3	68.4	72.5	66.3	62.0	59.2	55.0	(20)
	2%	DB	67.4	70.0	77.2	82.2	92.8	98.8	102.9	102.6	94.8	88.0	78.1	69.3	(21)
		MCWB	53.5	55.8	59.4	61.4	64.7	67.4	69.1	69.4	67.0	62.4	57.9	54.6	(22)
	5%	DB	64.2	66.6	73.5	77.7	87.6	94.0	98.6	98.3	89.7	84.4	73.8	65.6	(23)
		MCWB	52.7	55.2	58.4	60.9	64.1	67.6	69.2	69.8	67.2	62.4	57.1	54.2	(24)
	10%	DB	60.8	63.1	69.6	73.3	82.2	89.5	93.6	93.6	85.9	80.6	69.9	62.4	(25)
		MCWB	51.9	53.7	57.6	59.8	63.0	66.8	69.1	69.1	66.0	62.1	56.8	53.0	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.4	60.6	64.8	67.4	70.4	73.6	77.9	79.3	72.8	68.3	64.1	60.3	(27)
		MCDB	66.2	69.9	76.6	79.7	89.4	92.3	96.2	100.1	88.0	80.6	71.2	66.6	(28)
	2%	WB	56.2	58.1	62.1	64.5	67.7	71.4	74.4	74.7	70.6	66.4	61.7	57.9	(29)
		MCDB	63.4	66.7	72.8	77.1	83.5	90.6	92.6	92.3	85.9	78.5	70.9	63.7	(30)
	5%	WB	54.4	56.4	60.1	62.2	65.6	69.4	72.2	72.5	68.9	65.1	59.9	56.1	(31)
		MCDB	60.9	64.1	70.1	73.8	81.7	87.5	90.4	89.7	83.1	76.8	69.2	62.1	(32)
	10%	WB	52.7	54.6	58.1	60.3	63.9	67.8	70.4	70.9	67.6	63.8	58.1	54.4	(33)
		MCDB	58.9	61.7	67.4	71.0	79.6	85.1	88.7	88.2	81.3	75.1	66.9	60.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	17.5	18.8	20.3	21.2	23.2	25.8	27.1	26.6	22.8	20.8	18.1	17.7	(35)
		MCDBR	23.1	25.0	27.6	29.4	31.6	31.6	30.7	30.6	28.2	26.6	23.7	22.3	(36)
	5% WB	MCWBR	13.9	15.1	14.9	14.4	12.6	11.6	11.8	11.9	11.0	10.9	11.4	12.9	(37)
		MCWBR	19.3	21.7	23.8	25.5	27.2	28.1	27.8	27.5	24.1	21.8	19.0	18.3	(38)
(40) Clear-Sky Solar Irradiance	taub		0.297	0.331	0.381	0.389	0.419	0.453	0.497	0.478	0.423	0.384	0.335	0.307	(40)
		taud	2.537	2.430	2.274	2.274	2.210	2.105	1.966	2.033	2.220	2.339	2.476	2.538	(41)
	Ebn at Noon		293	291	283	285	276	265	253	256	266	268	275	282	(42)
		Edh at Noon	26	32	41	43	46	51	58	53	42	35	27	24	(43)
(44) All-Sky Solar Radiation	RadAvg		952	1217	1584	1887	2143	2439	2449	2208	1797	1351	1004	877	(44)
	RadStd		104	128	127	131	166	109	50	76	91	99	89	83	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	+2.34	-1.85	-2.02	N/A	N/A	+200	+288	(46)
(47)	Regional (6 neighbors)	N/A	N/A	N/A	+1.89	N/A	-0.86	N/A	N/A	N/A	+146	(47)

Nomenclature: See separate page